Worksheet #19

Week 10, Day 1: Clustering

Name: _____

NetID: _____

Date: _____

Guided checkpoints

For full credit, fill in the checkpoints. If you skip a checkpoint, you must include your own notes for full credit.

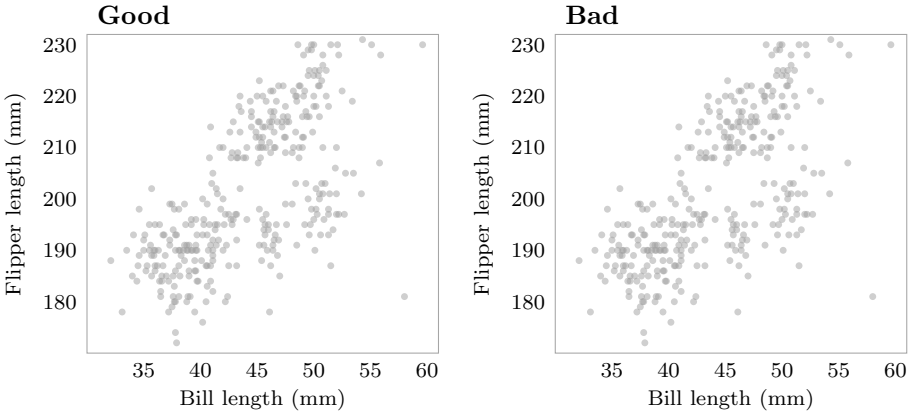
1. Supervised or Unsupervised?

Circle **S** (supervised) or **U** (unsupervised) for each task. If it is supervised, name the target variable.

Task	Type	Target (if supervised)
Predict a Cook County condo's sale price from its square footage and neighborhood.	S / U	
Group CTA stations whose weekday ridership looks alike.	S / U	
Predict a penguin's species from its bill and flipper measurements.	S / U	
Sort 4,500 movie reviews into groups without looking at their labels.	S / U	
Flag condo sales whose prices look unusual next to similar sales.	S / U	

2. Good Run, Bad Run

Both plots show the same 342 penguins from the slides. On the left, show where k-means ($K = 3$) ends up after a good run: draw the three final centroids ($\times$) and circle the three clusters. On the right, do the same for a bad run that gets stuck in a local optimum.



3. Choosing K

Why not pick the K with the lowest SSE?

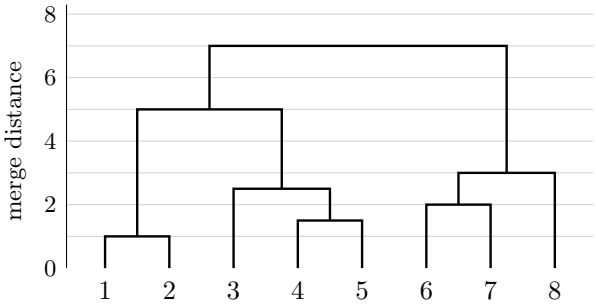
What is one method for choosing K?

4. Partitional or Hierarchical?

Circle **P** (partitional) or **H** (hierarchical) for the clustering each task needs.

Task	Type
Build a family tree of penguin species and their relatives.	P / H
Split a delivery area into 5 driver zones.	P / H
Organize songs into genres, subgenres, and sub-subgenres.	P / H
Group customers into 3 segments for a mailing.	P / H

5. Read a Dendrogram



Which two points are most similar? _____

Draw a horizontal line that cuts the tree into 2 clusters. List them:

How many clusters does a cut at height 4 make? _____